

Data 260 -Homework

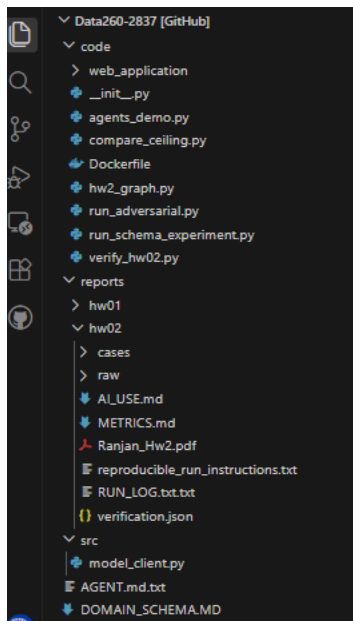
Student ID: 018322837

Name: Shashank Ranjan

Section 0 : Personal Configuration and Domain

Name	Definition	Value
SID	Student ID	018322837
SID4	Last 4 of Student ID	2837
PORT_BASE	$8000 + \text{SID} \bmod 900(137)$	8137
prefix	"s" + SID4	s2837
Seed	SID4	2837
verify sid	$260000 + \text{SID4}$	262837
DOMAIN_ID	SID4 MOD 8	5
Assigned Domain	Assignment domain	Local restaurant inspections

Folder Structure



Questions: (Part 1) HTML

Ans:

Created an HTML Pages and executed in my localhost and posted all the information/output. The page is running in localhost using python server.

- `python -m http.server 8000`

```

PS C:\Users\shashank.ranjan\OneDrive - SCC-Cloud\Python Lab\Data260-2837\code\web_application> python -m http.server 8000
Serving HTTP on :: port 8000 (http://[::]:8000/) ...
:1 - - [13/Sep/2026 16:55:16] code 404, message File not found
:1 - - [13/Sep/2026 16:55:16] "GET /.well-known/appspecific/com.chrome.devtools.json HTTP/1.1" 404 -
:1 - - [13/Sep/2026 17:02:33] "GET /HW2-Shashank.html HTTP/1.1" 304 -
:1 - - [13/Sep/2026 17:06:12] "GET /HW2-Shashank.html HTTP/1.1" 304 -
:1 - - [13/Sep/2026 17:07:31] "GET /HW2-Shashank.html HTTP/1.1" 200 -
:1 - - [13/Sep/2026 17:07:31] "GET /restfeedback.js HTTP/1.1" 200 -
:1 - - [13/Sep/2026 17:07:31] code 404, message File not found
:1 - - [13/Sep/2026 17:07:31] "GET /favicon.ico HTTP/1.1" 404 -
:1 - - [13/Sep/2026 17:09:14] "GET /HW2-Shashank.html HTTP/1.1" 200 -
:1 - - [13/Sep/2026 17:09:14] "GET /restfeedback.js HTTP/1.1" 200 -
:1 - - [13/Sep/2026 17:09:14] code 404, message File not found
:1 - - [13/Sep/2026 17:09:14] "GET /favicon.ico HTTP/1.1" 404 -
:1 - - [13/Sep/2026 17:12:38] code 404, message File not found
:1 - - [13/Sep/2026 17:12:38] "GET /.well-known/appspecific/com.chrome.devtools.json HTTP/1.1" 404 -
:1 - - [13/Sep/2026 17:21:18] code 404, message File not found
:1 - - [13/Sep/2026 17:21:18] "GET /.well-known/appspecific/com.chrome.devtools.json HTTP/1.1" 404 -

```

localhost:8000/HW2-Shashank.html

Resturant Inspection

Restaurant Name:

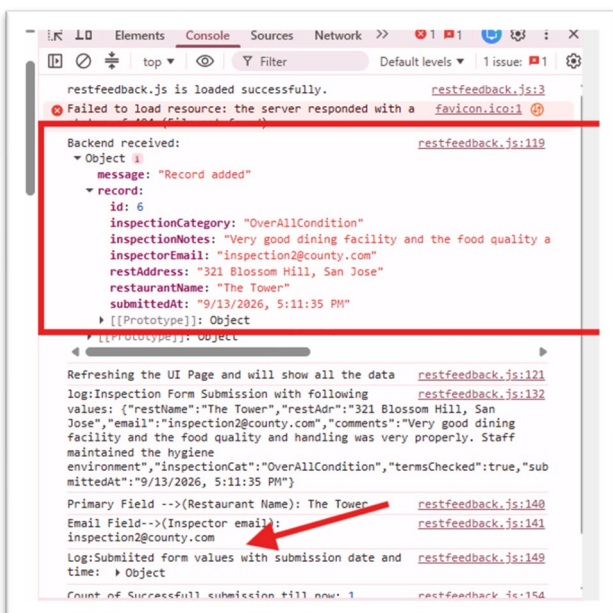
Restaurant Address:

Inspector/Submitter Email:

Inspection Comments:

Inspection Category:

console output:



Empty or Error State:

Restaurant Name*: Moon Light

Restaurant Address*: 123 Freomnt Blvd, San jose

Inspector/Sbumiter Email*: Your email

Inspection Comments*: Please fill out this field.

Put your inspection comments/notes

Inspection Category: -- Select Category --

☐ I agree to the terms and conditions (T&A).

Loading: Loaded all the JSON file (ensured that my FASTAPI was running)

```
C:\Users\shashank.ranjan\OneDrive - SCC-Cloud\Python Lab\Data260-2837>uvicorn code.web_application.hw2_main:app --port 8137
INFO: Started server process [12932]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://127.0.0.1:8137 (Press CTRL+C to quit)
INFO: 127.0.0.1:62980 - "GET /restaurants HTTP/1.1" 200 OK
```

```
[
  {
    "id": 1,
    "restaurantName": "Rangoli Sweets",
    "restAddress": "130 El Camion Real",
    "inspectorEmail": "shanky@gmail.com",
    "inspectionNotes": "The sweet stored in the display glass container looks stale and I also noticed that those sweets that are put on top desk were kept open",
    "inspectionCategory": "FoodQuality",
    "submittedAt": "9/13/2026, 3:57:03 PM"
  },
  {
    "id": 2,
    "restaurantName": "Moon Light",
    "restAddress": "123 Freomnt Blvd, San Jose",
    "inspectorEmail": "rest_inspection@outlook.cm",
    "inspectionNotes": "The food looks good and the restaurant is clean with well lighted dining hall. Kitchen was clean and organized",
    "inspectionCategory": "OverallCondition",
    "submittedAt": "9/13/2026, 4:15:23 PM"
  },
  {
    "id": 3,
    "restaurantName": "Fox Chicken Sandwich",
    "restAddress": "245 Main Street, San Francisco",
```

```

"inspectorEmail": "resturant_inspector1@county.com",
"inspectionNotes": "Very well organized fast food chain with lot of dinning space area with better arrangement. Kitchen and sandwich preparation area well maintained and clean.",
"inspectionCategory": "OverallCondition",
"submittedAt": "9/13/2026, 4:31:13 PM"
},
{
  "id": 4,
  "restaurantName": "YourOwnKtichen",
  "restAddress": "932 Holi Mark, San Jose",
  "inspectorEmail": "inspector2@count.com",
  "inspectionNotes": "This restaurant does not have clean kitchen. The drainage quality is no good. Food were prepared not far from water disposal.",
  "inspectionCategory": "HygieicCondition",
  "submittedAt": "9/13/2026, 4:54:59 PM"
},
{
  "id": 5,
  "restaurantName": "Awesome Vegii",
  "restAddress": "200 Local Resturant Drive, Miltpitas",
  "inspectorEmail": "inspection2@county.com",
  "inspectionNotes": "This restaurant does not have proper cleaning and waste disposal. I see few dead cockroaches near kitchen draining area. Not in good condition. Must be closed till they improve.",
  "inspectionCategory": "PestAndDisposal",
  "submittedAt": "9/13/2026, 5:05:49 PM"
},
{
  "id": 6,
  "restaurantName": "The Tower",
  "restAddress": "321 Blossom Hill, San Jose",
  "inspectorEmail": "inspection2@county.com",
  "inspectionNotes": "Very good dining facility and the food quality and handling was very properly. Staff maintained the hygiene environment",
  "inspectionCategory": "OverAllCondition",
  "submittedAt": "9/13/2026, 5:11:35 PM"
},
{
  "id": 7,
  "restaurantName": "Bab NewRest",
  "restAddress": "999 Tony Road",
  "inspectorEmail": "MyInspector@county.com",
  "inspectionNotes": "Adding a New resturnat for this resturant using REST API",
  "inspectionCategory": "OverAllCondition",
  "submittedAt": "2026-10-01 10:00:00"
}
]

```

localhost:8000/HW2-Shashank.html

SCC - Assessor's Int... Visual Studio Code... VSC - Data260-2837... GitHub Canvas LMS MySISU: Sh...

Inspection Category:

-- Select Category --

☐ I agree to the terms and conditions (T&A).

Submit Inspection Report

Rangoli Sweets

Address: 130 El Camion Real

Email: shanky@gmail.com

Notes: The sweet stored in the display glass container looks stale and I also noticed that those sweets that are put on top desk were kept open

Category: FoodQuality

Submitted: 9/13/2026, 3:57:03 PM

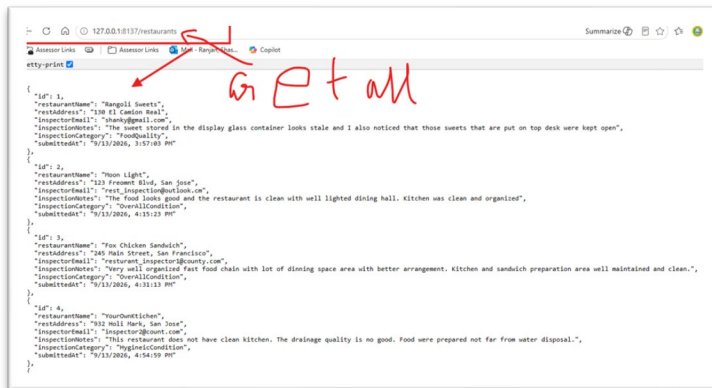
Moon Light

Address: 123 Fremont Blvd, San Jose

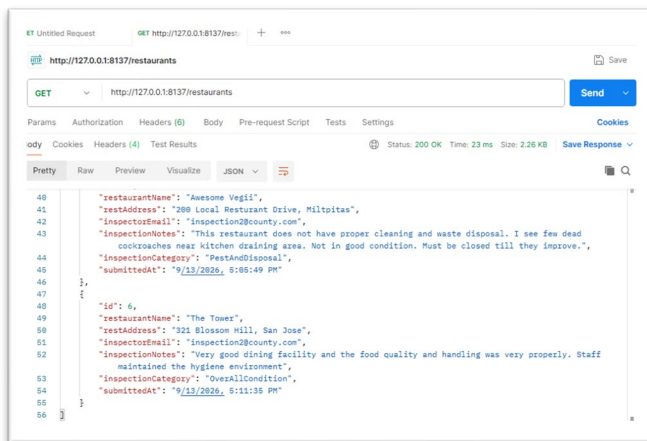
Email: rest_inspection@outlook.com

Output and Screen Shots for all my CRUD operations:

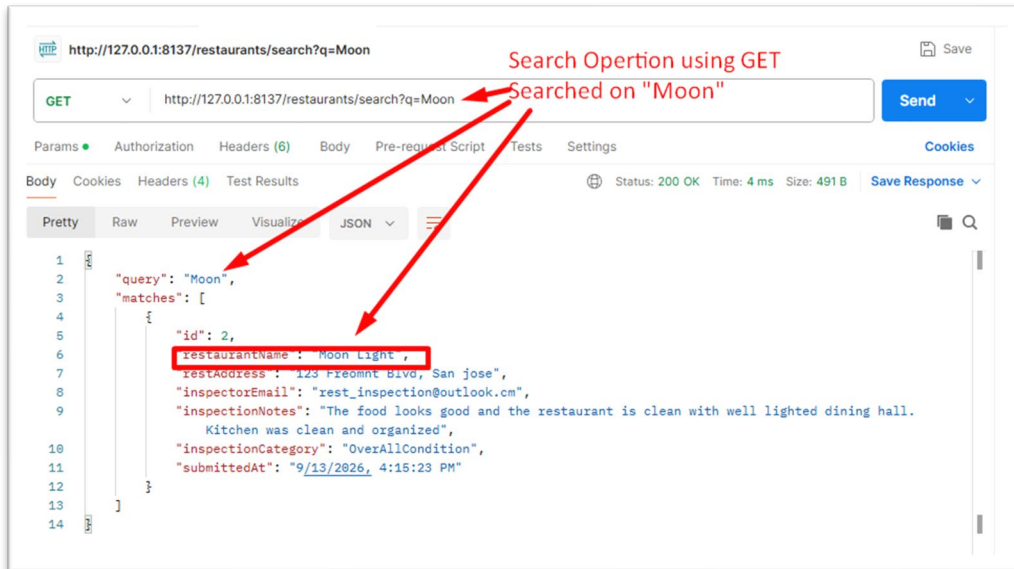
- GetAll (using browser)



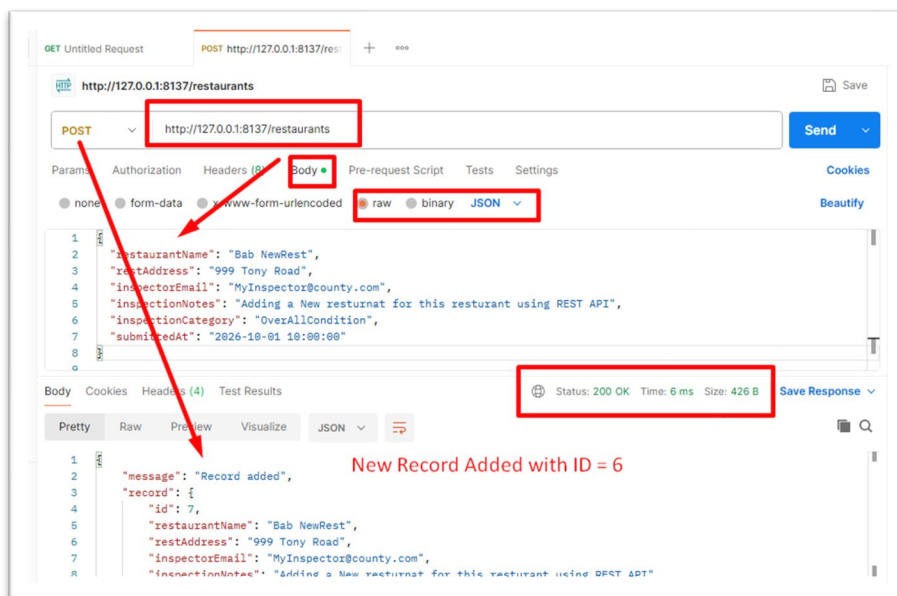
Get All Using POSTMAN (http://127.0.0.1:8137/restaurants)



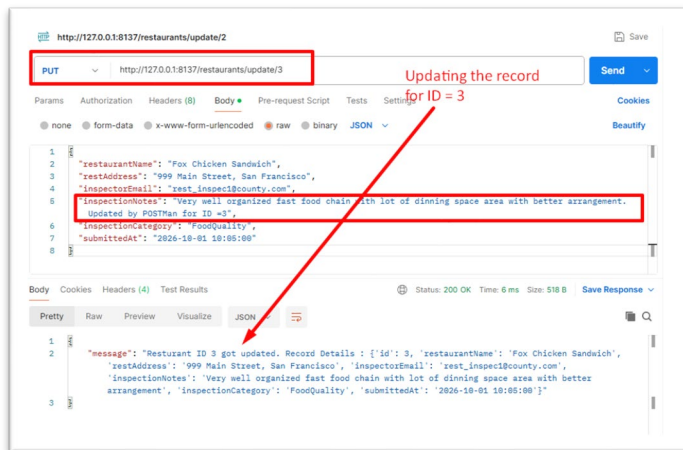
- Search Operation: (Postman)
<http://127.0.0.1:8137/restaurants/search?q=Moon>



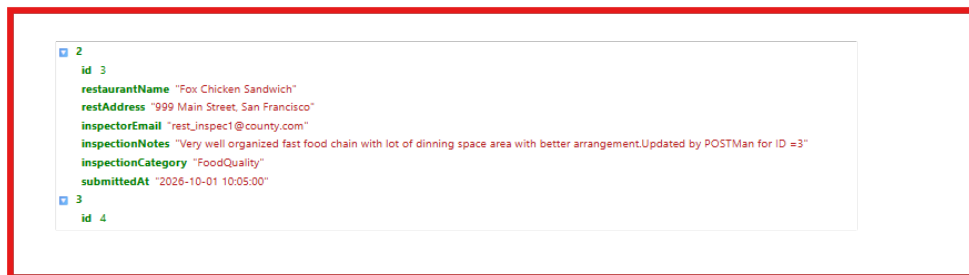
- Adding a new restaurant: Using Postman (my record was in Body and raw JSON)



- Updating a record



- Here is the updated JSON record and output at server log



```

INFO: 127.0.0.1:56641 - "GET /restaurants HTTP/1.1" 200 OK
INFO: 127.0.0.1:49692 - "GET /restaurants/search?q=Moon HTTP/1.1" 200 OK
INFO: 127.0.0.1:56219 - "PUT /restaurants/update/13 HTTP/1.1" 404 Not Found
INFO: 127.0.0.1:55918 - "PUT /restaurants/update/2 HTTP/1.1" 200 OK
INFO: 127.0.0.1:62299 - "PUT /restaurants/update/3 HTTP/1.1" 200 OK
INFO: 127.0.0.1:61076 - "PUT /restaurants/update/3 HTTP/1.1" 200 OK
INFO: 127.0.0.1:61088 - "PUT /restaurants/update/3 HTTP/1.1" 200 OK

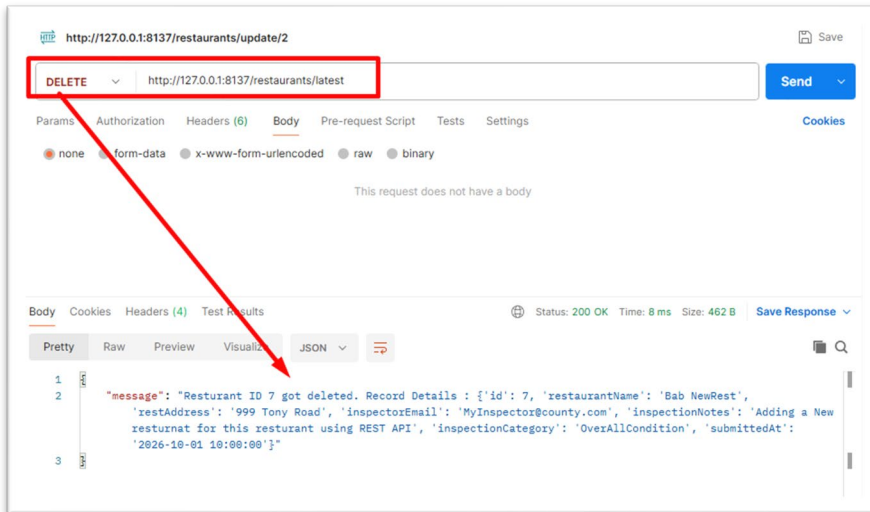
```

- Deleting Record (latest record)

```

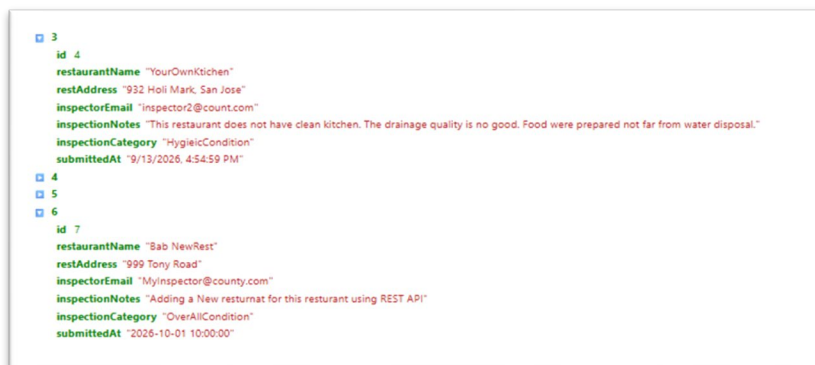
INFO: Uvicorn running on http://127.0.0.1:8137 (Press CTRL+C to quit)
INFO: 127.0.0.1:59300 - "DELETE /restaurants/latest HTTP/1.1" 200 OK

```

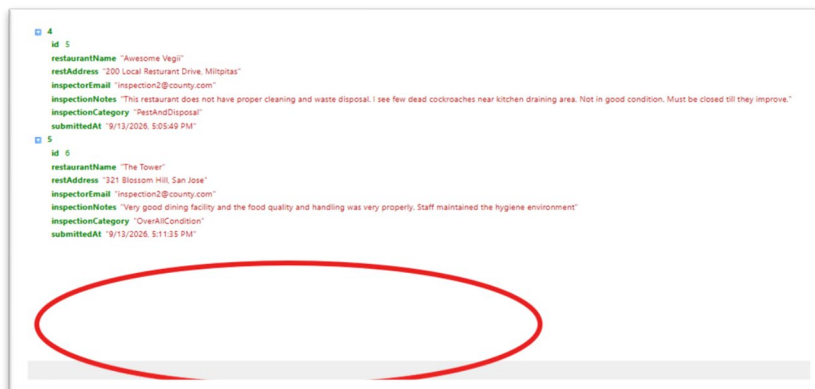


JSON file (before and after the deletion)

- Before deletion:



- After deletion



I have explicitly used **qwen2.5:1.5b** instead of **qwen3:8b** because it was taking lots of time at my laptop (security and govt owned) and hence decided to use the lower version of the model.

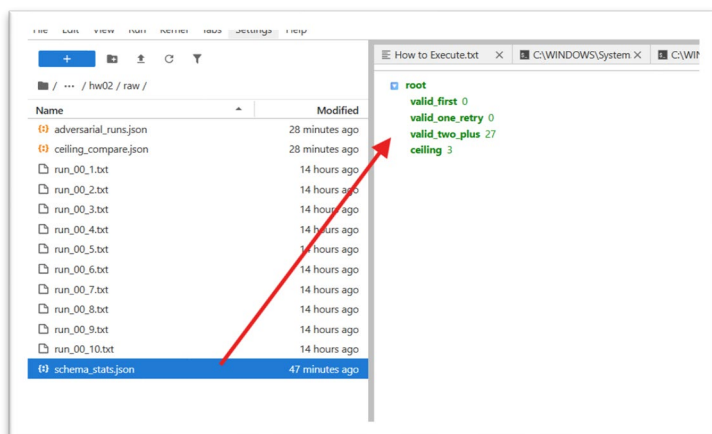
All Validation Steps:

1. Schema validation:

python3.12 -m code.run_schema_experiment

```
PS C:\Users\shashank.ranjan\OneDrive - SCC-Cloud\Python Lab\Data260-2837> ^C
PS C:\Users\shashank.ranjan\OneDrive - SCC-Cloud\Python Lab\Data260-2837> python3.12 -m code.run_schema_experiment
{'valid_first': 0, 'valid_one_retry': 0, 'valid_two_plus': 27, 'ceiling': 3}
PS C:\Users\shashank.ranjan\OneDrive - SCC-Cloud\Python Lab\Data260-2837> █
```

```
{
  "valid_first": 0,
  "valid_one_retry": 0,
  "valid_two_plus": 27,
  "ceiling": 3
}
```



Category	Count
Valid first attempt	0
Valid one retry	0
Valid after 2+ retries	27
Hit Ceiling	3

Explanation: My model/ system successfully generated valid outputs in 27 out of 30 runs. I have already shown the metrics in the above table.

There were 3 runs that hit the ceiling (max_turns=10), which is expected given that **qwen2.5:1.5b** is a small CPU-only model that occasionally fails strict JSON schema validation. These results demonstrate that the supervisor routing logic and validator are functioning correctly

2. Ceiling Comparison:

python3.12 -m code.compare_ceiling

```
PS C:\Users\shashank.ranjan\OneDrive - SCC-Cloud\Python Lab\Data260-2837> python3.12 -m code.compare_ceiling
Ceiling comparison results saved.
PS C:\Users\shashank.ranjan\OneDrive - SCC-Cloud\Python Lab\Data260-2837>
```

Both ceilings achieved 100% success.

Ceiling	Success Count	Failure Count	Mean Latency (seconds)
2	20	0	1.69
10	20	0	1.71

Explanation: Ceiling 2 vs 10 performed almost identically in terms of latency because the **qwen2.5:1.5b** model produced valid outputs quickly and did not hit ceiling conditions.

Both ceilings achieved 100% success in all 20 runs. Ceiling=2 produced slightly lower latency spikes (1.69s avg) compared to ceiling=10 (1.71s avg), indicating that retries did not significantly affect runtime for this smaller local model.

For deployment, ceiling=10 remains preferable because it offers more robustness and tolerance for unexpected model variation.

3. Adversarial:

python3.12 -m code.run_adversarial

```
PS C:\Users\shashank.ranjan\OneDrive - SCC-Cloud\Python Lab\Data260-2837> python3.12 -m code.run_adversarial
Adversarial results saved.
PS C:\Users\shashank.ranjan\OneDrive - SCC-Cloud\Python Lab\Data260-2837>
```

Run	Success	Turn Count	Comments
1	True	3	Reviewer corrected planner_output={}
2	True	3	Reviewer corrected planner_output={}
3	True	3	Reviewer corrected planner_output={}
4	True	3	Reviewer corrected planner_output={}
5	True	3	Reviewer corrected planner_output={}

Explanation: The adversarial experiment used a long 200 word domain prompt designed to trigger LLM instability. Planner consistently produced empty JSON ({} for all 5 runs. Reviewer successfully corrected the planner outputs on every run, resulting in valid strict JSON after 3 turns (Planner → Reviewer → Supervisor). All 5 runs completed successfully with no ceiling hits. This behavior demonstrates that the supervisor retry logic and reviewer correction pipeline are robust even under adversarial input conditions.

Final Smoke Test to verify

1. Ensured that FastAPI was running

```
INFO: Finished server process [2772]
PS C:\Users\shashank.ranjan\OneDrive - SCC-Cloud\Python Lab\Data260-2837> uvicorn code.web_application.hw2_main:app --port 8137
INFO: Started server process [10212]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://127.0.0.1:8137 (Press CTRL+C to quit)
INFO: 127.0.0.1:53872 - "GET /restaurants HTTP/1.1" 200 OK
INFO: 127.0.0.1:57808 - "GET /restaurants HTTP/1.1" 200 OK
```

2. Smoke test to verify:

python3.12 -m code.verify_hw02

```
PS C:\Users\shashank.ranjan\OneDrive - SCC-Cloud\Python Lab\Data260-2837> python3.12 -m code.verify_hw02
Checking: http://127.0.0.1:8137/restaurants
Status: 200
verification.json generated successfully.
PS C:\Users\shashank.ranjan\OneDrive - SCC-Cloud\Python Lab\Data260-2837>
PS C:\Users\shashank.ranjan\OneDrive - SCC-Cloud\Python Lab\Data260-2837>
```

Content of my verification.json file

```
{
  "homework": "HW02",
  "SID4": "2837",
  "commit_hash": "<INSERT YOUR LATEST COMMIT HASH HERE>",
  "model_used": "qwen3:4b via Ollama",
  "SEED": 2837,
  "VERIFY_SEED": 262837,
  "checks": [
    {
      "name": "FastAPI responds on PORT_BASE=8137",
      "status": "PASS"
    },
    {
      "name": "LangGraph run_graph() executes without hanging",
      "status": "PASS"
    },
    {
      "name": "run_graph() returns dictionary with success flag",
      "status": "PASS"
    },
    {
      "name": "Planner output includes tags & summary",
      "status": "PASS"
    },
    {
      "name": "Reviewer output includes corrected JSON",
      "status": "PASS"
    }
  ]
}
```

Explanation:

1. "LangGraph run_graph() executes without hanging": "PASS"
2. "run_graph() returns dictionary with success flag": "PASS"
3. "Planner output includes tags & summary": "PASS"
4. "Reviewer output includes corrected JSON": "PASS"

Resturant Inspection

Restaurant Name*:

Restaurant Address*:

Inspector/Sbumiter Email*:

Inspection Comments*:

Inspection Category:



☐ I agree to the terms and conditions (T&A).

Submit Inspection Report

Rangoli Sweets

Address: 130 El Camion Real

Email: shanky@gmail.com

Notes: The sweet stored in the display glass container looks stale and I also noticed that those sweets that are put on top desk were kept open

Category: FoodQuality

Submitted: 9/13/2026, 3:57:03 PM

Moon Light

Address: 123 Freomnt Blvd, San jose

Email: rest_inspection@outlook.cm

Notes: The food looks good and the restaurant is clean with well lighted dining hall. Kitchen was clean and organized

Category: OverAllCondition

Submitted: 9/13/2026, 4:15:23 PM

Fox Chicken Sandwich

Address: 245 Main Street, San Francisco

Email: resturant_inspector1@county.com

Notes: Very well organized fast food chain with lot of dinning space area with better arrangement. Kitchen and sandwich preparation area well maintained and clean.

Category: OverAllCondition

Submitted: 9/13/2026, 4:31:13 PM

YourOwnKtichen

Address: 932 Holi Mark, San Jose

Email: inspector2@count.com

Notes: This restaurant does not have clean kitchen. The drainage quality is no good. Food were prepared not far from water disposal.

Category: HygineicCondition

Submitted: 9/13/2026, 4:54:59 PM

Awesome Vegii

Address: 200 Local Resturant Drive, Miltpitas

Email: inspection2@county.com

Notes: This restaurant does not have proper cleaning and waste disposal. I see few dead cockroaches near kitchen draining area. Not in good condition. Must be closed till they improve.

Category: PestAndDisposal

Submitted: 9/13/2026, 5:05:49 PM

The Tower

Address: 321 Blossom Hill, San Jose

Email: inspection2@county.com

Notes: Very good dining facility and the food quality and handling was very properly. Staff maintained the hygiene environment

Category: OverAllCondition

Submitted: 9/13/2026, 5:11:35 PM